Causal Leverage Density: A Universal Framework for Semantic Information

Damian R. Sowinski[1,2], Silvia Holler[3], Mike Wong[4], Gary Segal[5], David Parkinson[6], Clément Vidal[7], James Henderson Cleaves[8], Pritvik Sinhadc[9], Anirudh Prabhu[4], Stuart Bartlett[9,2]

[1]University of Rochester, USA, [2]SETI Institute, USA, [3]University of Trento, Italy, [4]Carnegie Institution for Science, USA, [5]University of Queensland, Australia, [6]Korea Astronomy and Space Science Institute, Republic of Korea, [7]Vrije Universiteit Brussel, Belgium, [8]Howard University, USA, [9]California Institute of Technology, USA

sjbart@caltech.edu, drsowinski@gmail.com

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Abstract

Despite living in the information age, we often ponder: what does information mean? Shannon defined syntactic information using a simple measure applied to probability distributions. Syntactic information captures the degree of 'surprise' on the part of a message receiver, given some prior credences over the alphabet of symbols being transmitted through a channel; however it remains silent on semantics—the notion of meaning. This gap is especially critical for the artificial life community, which aims to understand and synthesize life-like processes where meaning and correlated function are essential. Existing approaches to semantic information are often domain-specific, tied either to linguistic contexts or to the viability of agents, limiting their generality. We thus introduce 'Causal Leverage Density' (CLD), a generalised approach to quantifying semantic information grounded in established concepts from statistical physics. CLD quantifies the influence of syntactic information by evaluating the effect of information-scrambling interventions on the future evolution of a system's phase space trajectories, which encapsulate all relevant degrees of freedom. This concept yields a universal approach to characterising meaning and semantic influence in any type of system, from physics to biology to machine learning. Crucially, by identifying systems where information has causal efficacy, CLD offers a robust tool for defining and detecting life as might be found elsewhere in the universe or created artificially on Earth.

Semantic information (SI), distinct from purely syntactic (Shannon) information, has been an important theme in many areas of Artificial Life (ALife), such as evaluating the performance of agents, robots and swarms, assessing natural and artificial cognitive systems, identifying emergent agents, and evaluating homeostasis in cybernetics (Brier, 2008). However, there are limitations to existing formalisms for SI, and in this paper we propose a generalised approach to SI called 'Causal Leverage Density' (CLD, first introduced by Bartlett (2024)). CLD measures the causal power of syntactic information by evaluating the impact of information-scrambling interventions on the future evolution of a system's phase space trajectories. This paper presents this new perspective as a foundation for technical developments and applications. We will first describe the background and previous works that endeavoured to quantify SI (Background), present our new concept (Semantic Information Theory and a Framework for Causal Leverage Density), discuss applications to key questions in astrobiology (Using Semantic Information to find Lyfe in Samples), and then discuss implications and future development steps (Conclusions).

Background

In today's world, one can perceive information as akin to a force of nature. It interacts with us every moment of our lives, moving across the globe and beyond at rates that would have seemed impossible only a few decades ago. Yet, despite these remarkable developments, it is still a challenge to define what information really means, and more so how it has come to hold such sway over the world around us. Information seems to have a life of its own (Scharf, 2022), despite it being designed (or naturally selected) to serve a purpose.

While we all have an intuitive sense of what information is, it can manifest in many nuanced forms such as data, measurements, knowledge, wisdom, predictive capacity etc. Some types of information seem useless, like the instruction manual for a cheap VHS player from the 1980s. Others are vitally important, such as the safety card for escaping from a plane when it has just crashed into the ocean. Certain information, like the enigmatic Voynich manuscript, might be potentially valuable though remain undeciphered. And then there's information that could become useful only in some distant or hypothetical future, such as unexpressed genes coding for proteins that might become crucial for survival (exapted) in novel, future conditions.

This state of mixed intuitive definitions of information was partially resolved by Claude Shannon, who introduced much of information theory (Shannon, 1948). The type of information defined in his famous works is typically described as syntactic, pertaining to symbols and the probabilities of their occurrence. This stands in contrast to SI, which carries meaning and, as a result, has the capacity to affect the world or influence the behaviour of agents within it.

The semantic–syntactic division was defined by Shannon's inaugural analysis of communication channels: focus-

ing solely on the probability of this or that string of symbols conveyed from source to receiver sweeps under the rug the question of any meaning conveyed during communication (Shannon, 1948).

Efforts to bridge the semantic–syntactic divide have largely relied on analytical methods closely linked to language, particularly higher order logics, which offer increasingly refined tools for capturing subtle linguistic features such as predicates and modalities. Beginning in 1953, Bar-Hillel and Rudolf Carnap launched such a program to study SI, quantifying meaning in messages through their relationship to "true descriptions of the world" (Bar-Hillel and Carnap, 1953; Dretske, 1981; Fodor, 1990; Stampe, 1977).

Fast forwarding towards the modern age, the field of causal inference has been revolutionised by Judea Pearl and collaborators (Pearl and Mackenzie, 2018), who introduced a formal framework for quantifying causal power (beyond simply correlation) of specific actions (e.g., the administering of a drug) established through interventional experiments, the primary example being randomized controlled trials. In this approach, causal factors are added or removed systematically (these are the interventions), and the system is re-assessed to establish whether a different outcome occurs (e.g., a disease resolving due to the drug). The introduction of a "calculus of interventions" revolutionized drug trials (Greenland, 1999) and generated a possible framework for understanding how consciousness emerges from nerve complexity (Albantakis et al., 2024).

In pursuit of measurements of the reproductive success of digital organisms, Adami (1998) introduced a formalism for computing the fitness of those agents. This formalism quantified how much information in their genomes contributes to replicative success, including performance on externally defined tasks that boost reproductive rate. Additionally, the approach of Standish (1999) measures the complexity of digital organisms by quantifying the rarity of genotypes that yield the same phenotype, where phenotypes are defined by intrinsic ecological interactions within the digital evolving system. These approaches are similar in spirit to the 'functional information' introduced later by Hazen et al. (2007) and generalised in Wong et al. (2023). This metric measures the rarity of functional sequences or configurations in a given combinatorial space of potentially functional entities.

Approaches grounded in non-equilibrium thermodynamics have also emerged recently, proposing that meaningful information is inextricably linked to viability (Kolchinsky and Wolpert, 2018; Piñero et al., 2024; Piñero et al., 2024) (note that we henceforth refer to the work of Kolchinsky and Wolpert (2018) as 'KW18'). The namesake of these *viability*-based approaches (VBA) is defined as the degree to which a living system keeps itself far from thermodynamic equilibrium. Hence in living systems—such as cells or organisms, viability is closely tied to the fundamental goal of surviving. Given that organisms consistently engage in behaviours aimed at preserving life, it is reasonable to infer that viability holds intrinsic meaning for them. Over time, the dynamic interactions between organisms and their environments give rise to co-evolved correlations, effectively shaping environments into niches that enhance the chances of survival. One of the central aims of VBA to semantic information theory (SIT) is to quantify the exact contribution of these correlations to an organism's viability-offering a measurable proxy for their semantic value or meaningfulness.

Doing so is no easy task: one first measures baseline agent–environment correlations alongside expected viability, then systematically 'scrambles' the correlations to identify which are crucial to survival. Small-scale scrambling typically has minimal or no impact (unless a critically important piece of information is lost at this stage, such as an essential gene, for example), but as more correlations are disrupted, an organism reaches a tipping point. Once a critical number of correlations have been scrambled, every further disrupted bit is expected to have a catastrophic impact on the organism's viability. In the context of VBA, this threshold marks the boundary between syntactic and SI (at least from the perspective of the organism or agent) (Sowinski et al., 2023).

When performing information scrambling, we are essentially creating counterfactual versions of a system—parallel worlds in some sense. Central to such counterfactual formulations, including both Giulio Tononi's Integrated Information Theory (IIT) (Albantakis et al., 2023) and the SIT of KW18, are searches over purviews of possible interventions. In analogy to Pearl's 'do' operator (Pearl, 2019), an intervention involves fixing the values of particular degrees of freedom (DoF); searches are therefore over the power set of system substates, making applying these formulations not just infeasible, but computationally impossible to apply directly to realistic systems. Applying IIT directly to *C. Elegans*, for example, whose tiny nervous structure of 302 neurons has been completely mapped, requires analyzing approximately 10^{91} states, a number nearly 100 billion times larger than the total number of particles in the observable universe (Bentley et al., 2016).

The computational complexity of these formulations is a crucial hurdle to overcome. In the case of IIT, Max Tegmark has proposed different integration measures (Tegmark, 2016) that behave in a trivial way on large volumes of subsets of elements, allowing searches to focus on particular regions akin to MCMC sampling. It is unclear whether such changes should be considered approximations, or if they radically change the predictions of IIT in unforeseen ways.

Closer to home, there are now several cases of systems with gargantuan state spaces which have been shown amenable to SIT by one of the authors. These sys-

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tems include foraging in replenishing environments (Sowinski et al., 2023), (de/)synchronisation control across network topologies (Sowinski et al., 2024a), and maintenance of planetary disequilibria by ecosystems (Sowinski et al., 2024b). Success in these examples came from case-dependent coarse-graining of the state space and methodical selection of interventions, resulting in tractable systems that were found to preserve the predictions of VBA to SIT. These successes in simplifying the search over interventions are informed by VBA, but are not wedded to it, opening the door to more general approaches to SIT.

While viability may be an important metric—since we, as organisms, are imbued with the drive to survive—there are many example situations where the degree to which some information can influence the future has nothing to do with viability or where viability cannot capture all of the significance of that information. The tightening of a communication channel or the establishment (or destruction) of a correlation may not fundamentally harm an organism, but may force it to change in some crucial way. What is the semantic content of such information without the metric of viability to guide us? As humans, we encounter many events in our lives where a certain piece of information, perhaps stochastically impressed upon us like a barely overheard conversation at a pub, can alter the path of our lives dramatically without having a negative or positive impact on our overall viability. For example, one hears about a job opening at a local firm and takes the brave step of calling the CEO, lands the job, meets their future spouse and lives happily ever after. A counterfactual timeline, wherein the information about the dream job had not reached the ears of our protagonist, would be more or less equivalent in terms of the expected lifespan of said character. And yet, the semantic content of that little, barely heard slip of conversation screams volumes.

Thus, CLD aims to capture other possible senses of meaning that could be relevant to a given system, with viability being a special case wherein meaning and viability are commensurate. One of our key motivations in seeking a more general framework for SI is the search for extraterrestrial life. We have previously suggested the definitions of life and habitability should be dramatically broadened so as not to miss examples of life as we do not know it—that is, 'lyfe' that is markedly different in composition from terrestrial life (Bartlett and Wong, 2020; Wong et al., 2022). We proposed that an essential criterion (among others) for a system to be counted as living is whether it has a deep relationship with information (whether it is capable of learning).

Clearly, a system capable of learning will have repertoires of syntactic information, and communication channels with (sensitivity to) the outside world. This information will also carry a semantic content: some of it will be more meaningful to the organism (e.g., where it can find free energy sources), and some of it will be less meaningful (e.g., correlations in the momenta of its boundary molecules with random Brownian fluctuations). Therefore, SI has manifest utility in astrobiology: when searching for new forms of life, the extent to which a system contains and utilizes SI should correlate with the likelihood that that system is alive. Hence, SI could be used as a biosignature, primarily in the realm of in situ sample analysis, where standard tests such as genetic sequencing, searching for canonical biomolecules, and morpholohical tests could be futile for novel forms of lyfe.

The viability framework set forth by KW18 is amenable to these types of life detection tests. One could in principle perform a range of information scrambling interventions on a putative alien living sample, and measure the degree to which a target subsystem suffers a viability loss. CLD offers a complementary approach, to establish whether a target system contains information which is meaningful in *any* sense, not just in the sense of affecting whether it survives or dies. In addition, applying VBA to *in situ* life detection would likely entail damaging or killing putative alien life forms, which may be highly undesirable from an ethical standpoint.

The approach of CLD, described in the following section, focuses instead on the influence of syntactic information on the phase space trajectories of a system. We note that a framework for measuring causality and SI has not, to the proposers knowledge, thus far been presented through this lens of phase space trajectories, as proposed for CLD (Bartlett, 2024). Note that our approach is broader than classic perturbative approaches to dynamical systems (including the Lyapunov exponent), wherein one simply assess whether initially close trajectories diverge.

This concludes our very brief survey of key contributions to SI. In the next section we briefly review elemental aspects of information theory, discuss VBA to SIT, and then turn to our proposed causal leverage approach (CLA) to SI, which is grounded in the defining starting point of statistical physics: the phase space representation of the state of a system.

Semantic Information Theory and a Framework for Causal Leverage Density

Information, as introduced by Shannon Shannon (1948), is what manages uncertainty — *gaining syntactic information* is equivalent to *reducing uncertainty*. Shannon showed that the uncertainty about a random variable X takes the form

$$H(X) = -\sum_x p_X(x) \log_2 p_X(x), \tag{1}$$

naming this quantity the *entropy*. Here $p_X(x)$ is the probability that X takes on the particular value x, the sum is over all possible values of x, and the logarithm is base two: the units of entropy are bits. Under the $\mathcal{Q}$ interpretation, Equation 1 is equal to the minimum number of YES/NO questions needed to infer unambiguously that $X = x$ (Gleiser and Sowinski, 2018; Sowinski et al., 2025).

Since correlations between two parts of a system imply that learning the state of one part reduces uncertainty in

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the state of the other, they quantify information. Shannon framed this in the context of communication, relating the correlation between sent and received messages with the rate at which those messages can be sent in his revolutionary noiseless and noisy channel coding theorems. Key to his setup was the notion of *mutual information*: given two random variables X and Y (such as the sent and received messages in a communication channel) it is

$$I(X:Y) = H(X) + H(Y) - H(X,Y). \tag{2}$$

Again, under $\mathcal{Q}$, it is the reduction in the number of questions that one must ask to determine $X = x$ and $Y = y$ as a pair rather than individually. Such a reduction is only possible if learning one of the two tells you something about the other—hence the *mutual* adjective (Sowinski, 2016).

The beauty of Shannon's setup is its complete generality, leaving open the nature of the communication channel, the sender and receiver, and any possible meaning the messages might have. For example, imagine yourself walking through a garden in spring. You look at a bed of flowers, taking in their colors, shapes and scents. You don't touch the roses, but pluck a lilac from its tree to carry with you. With a generous interpretation of communication, let's view this in Shannon's terms. The light bouncing off the flowers, the odors released by them—these are all part of a message continuously sent by the garden. The message is the state of the garden, a message that can be disrupted by rain or fog, disruptions causing you, the receiver of the message, to look longer to discern what you see. When you receive the message, it informs your decisions, and you act accordingly, interactions that further correlate your state with that of the garden. Information is exchanged, processed, and stored as you commune with the garden.

Inferring the state of the garden through sights and smells showcases the syntactic aspects of information; it is your actions with respect to this inference where semantics hides. Why did you not touch the rose? Why did you pluck the lilac? If you had seen them in opposite order, would you have not touched the lilac but plucked the rose? Avoiding a bloody finger is likely behind a decision to not pluck the rose; that would not change had you encountered the flora in different order. The information about flower type represents a temporal correlation between the present state of the garden and your predicted future states, modulating your actions. Information is semantic if it influences decision making to emphasize particular classes of future states; the commonality within that class providing the meaning.

For example, a VBA to SIT uses the semantics of viability, or health Sowinski et al. (2023). Letting A and E denote the random variables representing your state (the agent) and that of the environment (the flowers in the garden), respectively, $V_\eta(A|E)$ is your viability in the context of the environment, while the correlation between the two of you is $I_\eta(A:E)$, as described by Equation 2. The effect the latter has on the former is found by considering counterfactual worlds in which correlations are removed. η corresponds to the level of information scrambling, ranging from $\eta = 0$, corresponding to none, to $\eta = 1$, full scrambling (resulting in $I_1(A:E) = 0$). For every possible way of changing the system, one measures and plots the point (I_η, V_η) in the information–viability plane, resulting in region called a *viability curve.* See Figure 1 for a synopsis.

A VBA does not capture all the intuitive notions of meaning. Wrapped up in seeing the rose is a lot of information about color, size, shape, etc. Most of this is considered syntactic in VBA, since, for example, in a counterfactual world in which they are a different color (either by colorblindness, pigmentation, etc.), your decision to not pluck one would likely not change and not affect your health. This reflects the general observation that the viability curve contains a plateau—an upper bound on viability saturated (or nearly so) for a great many η. However, in a counterfactual world where you did not know that roses have thorns, you might just pluck one causing your finger to suffer. The η's scrambling this information map to regions below the semantic threshold in the information–viability plane—as can be seen in the viability curve in the lower left panel of Figure 1. Any scrambling that results in non-saturated viability implies that the correlations are semantic. In VBA, "thorniness" is meaningful, yet almost everything else about the rose is not.

What about the lilac? If your decision to pluck it is based on your love of its deep lavender colour, then in any counterfactual world in which the colour of the lilac was different you would not pluck it. In either case, your health would be unaffected. Clearly the colour of the lilac is meaningful to you, yet according to VBA the colour is meaningless. This limit of the approach is precisely what a CLA addresses, to capture a more general notion of meaning than health: change. There are many cases where an organism or population could contain information that is meaningful, but rather than causing a drop in viability (e.g., death, a bloody finger), the loss of that information could simply change the living system. The information loss could cause it to adapt to the perturbation or adopt a new modality, such as prokaryotes going into stationary phase if they no longer sense their external temperature. Rather than focusing on viability, a CLA to SIT attends to the more general notion of *influence*. Assessing the effectiveness of interventions (scrambling syntactic information) on the phase/configurational space trajectories of a system (influence on the future) is the promise of CLD (see Figure 2).

Specifically, the CLA assumes a general scenario of stochastic dynamical systems where there is uncertainty in the precise values of the defining DoF. States, Ψ, are therefore represented by vectors in a Hilbert space (though the formalism does not preclude quantum effects, we restrict ourselves to classical systems i.e. real, not complex, Hilbert spaces). These are dependent on time, t, and we write the

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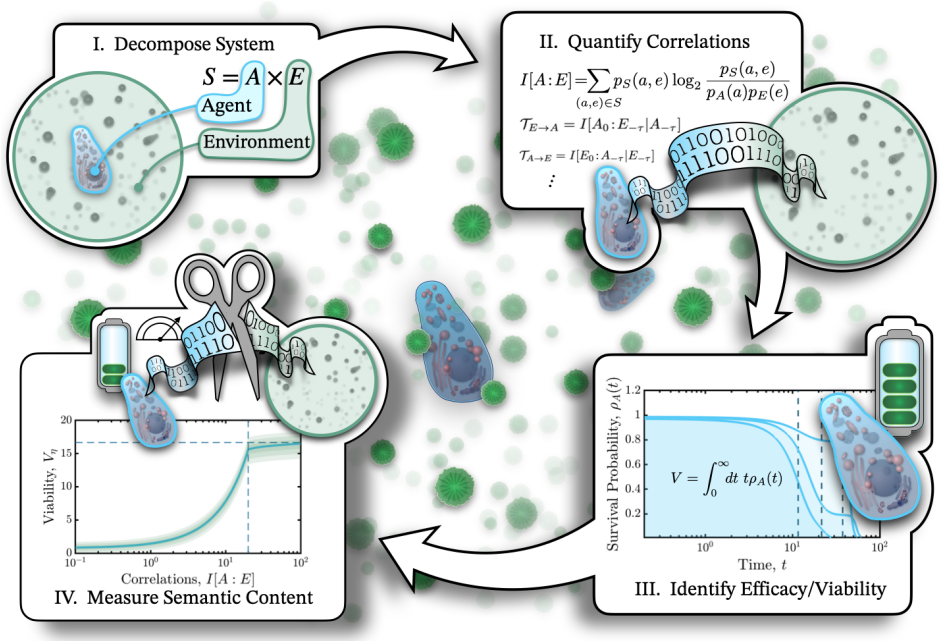

Figure 1: VBA and CLA to SIT consist of four general steps. The first is to identify the coarse-grained DoF describing a system, and how they characterize the agent and its environment. The next step is to quantify the correlations using information theoretic measures such as the mutual information described in the text. This is followed in VBA by measuring the viability of the agent in the environment using the agent's coarse-grained DoF. In the CLA a more general measure of efficacy needs to be developed. Lastly, scrambling over all possible subsets of DoF reveals the viability or efficacy curves as described in the text. Key features of the viability curve in VBA include the syntactic plateau and the semantic threshold. It is not yet clear if there will be analogues to these structures in the CLA.

vector representation of the state as $|\Psi, t\rangle$. A particular agent and environment, a and e, has a vector representation, $|a, e\rangle$. The joint probability of finding $A = a$ and $E = e$ at time t is $p_{AE}(a, e; t) = \langle a, e|\Psi, t\rangle$. The state of the system evolves continuously via a master equation,

$$\frac{d}{dt}|\Psi, t\rangle = \boldsymbol{R}|\Psi, t\rangle, \tag{3}$$

where $\boldsymbol{R}$ is the transition rate matrix for the coarse-grained system. For systems having continuous DoF with finite moments, master Equation 3 further reduces to a Fokker-Planck equation. In either case, the resulting evolution describes what the system does naturally, without interventions. Scrambling syntactic information through interventions (e.g., knocking out a gene in a cell) at some time t' is done by projecting out the correlations belonging to the corresponding subspace, $|\Psi, t\rangle \mapsto |\Psi', t\rangle = \boldsymbol{\Pi}(t')|\Psi, t\rangle$, leading to an intervened-upon distribution, $\hat{p}(a, e; t') = \langle a, e; t'|\Psi', t\rangle \neq p(a, e; t')$. Post intervention, the original and counterfactual trajectories, $|\Psi, t\rangle$ and $|\Psi', t\rangle$ respectively, differ. Whether in the future they diverge or recoalesce is what CLD should quantify and encapsulate.

The key to successfully applying the CLA to SIT will be engineering a measure quantifying this divergence of trajectories. What properties should this measure have? In general, all trajectories eventually converge 'at the heat death of the universe'. On more measurable timescales, all living things, as far as we know, eventually die. CLD must capture more than the simple binary property of convergence or divergence of trajectories, but be able to assess scenarios such as those in Figure 2, and more. Any time the trajectories converge on a relatively short timescale, the CLD should be low, since the change induced by the intervention did not last long. If they diverge, it should be high. However, considerations of symmetry become important as well. For example, in some systems the original and intervened trajectories may diffuse, but the latter significantly less than the former (for example, a system that has become more strongly homeostatic after intervention). Alternatively, an intervention may induce the exact opposite (e.g., losing key information for homeostasis, causing a breakdown of system stability). Should the symmetry in these two scenarios under exchange of original and intervened upon trajectories be broken by CLD?

The broadest interpretation of CLD is an integrated measure of phase space divergence over all time after the intervention (until the system has 'equilibrated' and the two sets of trajectories are essentially indistinguishable). In practice this will generally be impossible, since we cannot observe a

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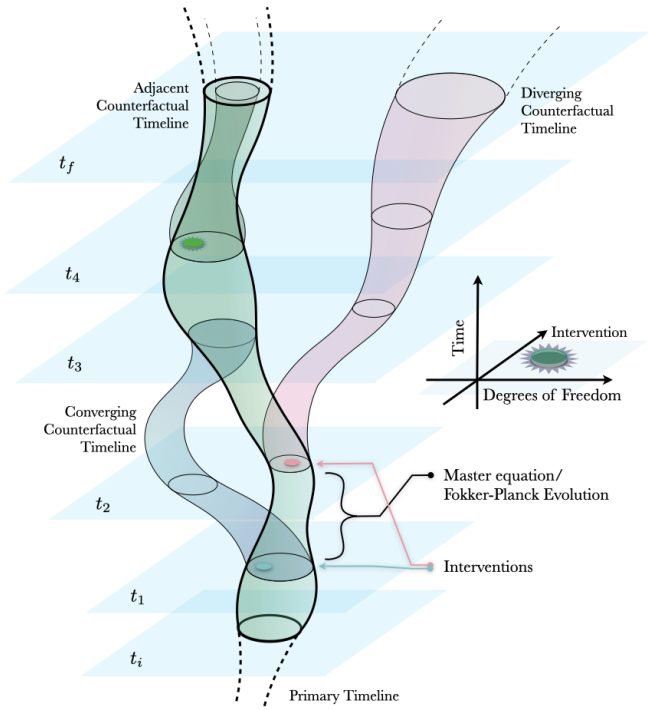

Figure 2: "Phase Spacetime" diagram of a system's DoF, with time flowing in the vertical direction between some initial (t_i) and final times (t_f). The tubes represent the probability distribution functions over these DoF. Counterfactual histories are generated by interventions to the primary history (in bold) at specific instances, after which the counterfactual history continues evolving. Several scenarios are shown. (Blue) This intervened timeline initially diverges from the primary one, but eventually converges back to it becoming indistinguishable. (Pink) This timeline continues diverging from the primary timeline. (Green) This timeline remains near yet distinct from the primary timeline.

system long enough to see the equivalent of the heat death of the universe. Thus the CLA must encompass several complementary measures of CLD:

1. The instantaneous divergence rate of trajectories, somewhat similar to the Lyapunov exponent (Young, 2013),

$$\chi_1(t) = F_1(|\Psi, t\rangle, |\Psi', t\rangle). \tag{4}$$

2. The maximum observed divergence over the time that the system is monitored (e.g., the duration of the experiment). So after choosing a time interval starting at t_1 and going until $t_2 > t_1$,

$$\chi_2(t_1, t_2) = \max_{t \in [t_1, t_2]} F_2(|\Psi, t\rangle, |\Psi', t\rangle) \tag{5}$$

3. The average divergence over the same time interval,

$$\chi_3(t_1, t_2) = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} dt \;\; F_3(|\Psi, t\rangle, |\Psi', t\rangle). \tag{6}$$

Here the three F_i represent functionals of the original and intervened on system states.

In terms of specifics for these functionals, the Jensen–Shannon (JS) divergence (divided by the bits scrambled by the intervention) was proposed in Bartlett (2024),

$$F_i(p, \hat{p}) = \frac{JS(p||\hat{p})}{\Omega_{scr}} \tag{7}$$

as this is a natural, symmetric measure for assessing differences in probability distributions, with an interesting $\mathcal{Q}$ interpretation: Consider using a fair coin to choose either the original or intervened system, then measuring that system. The JS divergence is the mutual information between that random measurement and the random choice you made Klir and Wierman (1999). Hence measuring a value that can discern whether it came from the original or from the intervened trajectory corresponds to a large CLD. Nonetheless, the JS divergence is not the only choice available, and other options need to be compared and contrasted. It is worth noting that many, if not all, continuous divergence measures reduce to the Fisher–Rao distance on the statistical manifold in the limit of infinitesimally small interventions Rao (2009); Nielsen (2020). From this we conjecture a deep relationship between CLD and information geometry, one which will be fleshed out in future work.

This concludes our summary of the CLD framework. Current and future endeavours will firmly establish the mathematical details of the metric, alongside various tests in both experimental and simulated systems. Many of these tests will establish the utility of CLD for detecting life in astrobiological contexts, which we detail in the next section.

Using Semantic Information to find Lyfe in Samples

Living systems, such as single cells, exhibit high SI due to their ability to detect and interpret environmental stimuli and to initiate adaptive responses that promote survival. In contrast, non-living systems may exhibit considerable physical or dynamical complexity but generally lack the goal-directed functionality and context-dependent information processing inherent to biological systems. Fire, for example, is a non-equilibrium dissipative structure which can propagate rapidly, mimicking a form of reproduction. However, there is no sense in which we could erase information or communication channels within a fire and have any real impact on its viability (this is in some sense why they are so destructive, as compared to say, viruses, which can be countered with vaccines that contain the appropriate SI). We can rearrange the molecules in a fire in almost every conceivable way, but its identity as a fire would be unaffected, provided one does not remove all the external oxygen or the locally high temperature. Consequently, while fire can spread and maintain itself as a dissipative system, it carries little SI, it

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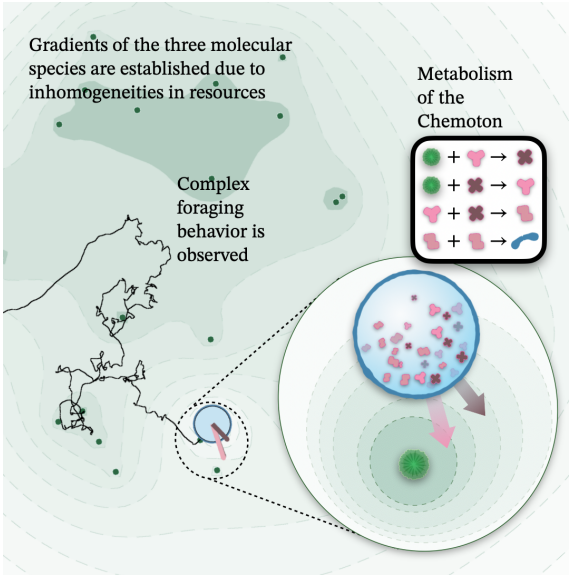

Figure 3: A chemoton (blue circle) with a metabolism made of three molecules (pink shapes) and a membrane (dark blue) fueled by resources (green). Green contours represent gradients of these resources. Reactions (rightmost inset panel) with the resource create a gradient in the metabolic molecules, with one species responsible for rebuilding the decaying membrane which manifests as translational motion (black trajectory).

simply exhibits spatial correlations with external gradients of temperature, oxygen, fuel, etc.

As humanity extends its exploration deeper into the cosmos, we may encounter dissipative structures far more complex than fire that exhibit characteristics commonly associated with life, such as individuality, replication, and motility. However, if these systems are of a very different composition to known life, distinguishing them from abiotic phenomena presents a significant challenge. Trying to identify life through other means, such as behaviour, could also be problematic. Many abiotic systems display complex behaviours that could be mistaken for life (Barge et al., 2022; Egbert et al., 2010). Simple models of chemotons, for example, can lead to strikingly life-like behaviours (Gánti, 2003); see Figure 3, and animations of a thermal reaction-diffusion model (Bartlett and Bullock, 2015).

At what threshold does the complexity of a system's behavioural repertoire warrant its classification as living? SIT offers a pathway to address this fundamental question. Observing responses to perturbations, and going beyond passive observation, informational interventions could inform the level to which a putative alien cell is alive, by observing its responses to perturbations.

If it was suspected that an alien cell senses its environment and communicates with it, or contains information-bearing structures, an information scrambling experiment could be performed, followed by an assessment of the cell's viability. One could additionally analyse whether the system diverges in its state space. The magnitude and form of that divergence could inform estimates of CLD for that system.

Living cells exhibit correlations (information exchanges) with their environments in many ways. In much biological research information storage molecules such as DNA are emphasised. These molecules directly modulate the phenotype of the cell, and depending on phenotypic variations, a cell communicates and interacts with its environment. Apart from DNA, cells also store information in other ways, such as epigenetic markers, concentrations of key biomolecules, and activation states of ligands. In principle, any of these correlations (syntactic information) between cell and environment could also be semantic, since they could be meaningful to the cell, and modify its phenotype and interaction with the outside world.

VBA to SIT propose that intervening upon (disrupting) any such information exchange between cell and environment can highlight the degree to which that information is meaningful, similarly to the effects that a gene knockout (KO) could have. We consider KO a specific case of SIT applied to a biological scenario, and propose that CLA could be developed to generalise and operationalise the causal influence of information. Note that both information flows (e.g., molecular sensing) and information-bearing structures (e.g., DNA) constitute correlations between agent and environment, and interventions on any such correlations could be used to test for SI.

We propose a series of experiments that could be conducted, starting with simple abiotic structures, progressing to real prokaryotic cells, and simple eukaryotic cells, to develop SI detection protocols. This set of systems would permit the establishment of a representative scale of SI values.

Life as we don't yet know it, could exploit very different mechanisms for information storage. Hence we suggest the development of general experimental interventions to be applied to putative alien cells that could assess whether there are communication channels or information-bearing structures within them. Information exchange can be disrupted by perturbing the boundary between cell structure and environment, and information-bearing molecules could be disrupted by scrambling their monomer sequences.

In order to benchmark a representative set of SI values, and understand the degree to which abiotic structures can also exhibit SI, the first type of cellular entities that could be studied is artificial synthetic cells, specifically lipid vesicles (Deamer, 2017; Luisi et al., 1999; Tessera, 2009) and coacervates (Jia and Fraccia, 2020; Matsuo and Kurihara, 2021; Stefano and Kream, 2025). In terms of assessing SI, one could apply interventions to disrupt any potential information flows between the structure and its environment. These interventions would include varying external param-

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eters such as salt and nutrient concentration and temperature. Since these simple protocells are not alive, we expect that there will be limited or no sensory communication channels to the environment, no molecules bearing information to scramble, or if there are, scrambling such information should not impact their viability. The majority of interventions upon the system will not prevent these protocells from reforming again, or producing a change in the type of protocells that do emerge. Hence we anticipate the viability-based SI to be low. However, CLD can account for SI when that information is unrelated to viability, ideally suited to the situation that might occur if an intervention perturbs a set of protocells such that the original set morph into a different type, shape or population.

More complex protocells, such as those which show a reaction to variations in their environment, might be expected to have finite SI content. A simple example is the well-studied chemotactic oil droplet, capable of moving in response to external concentration gradients, analogous to a single-celled organism (Čejková et al., 2017; Holler et al., 2018; Holler and Hanczyc, 2020). In fact, these are essentially experimental versions of the chemoton discussed in Figure 3. The droplets exchange information with their environments, due to local surface tension variations, which causes the droplets to move in a specific direction. Therefore, this protocell behaviour could be affected by information-scrambling interventions, such as re-arranging boundary molecules. This would erase correlations between the concentration gradients immediately external to the protocell, and gradients of surface tension in the protocell. By the VBA definition of SI, this system would have low SI, because its viability is not affected by interventions that disrupt local concentration patterns. However, it would have a finite CLD, because its behaviour would be transiently affected by certain interventions. This example highlights the importance of developing CLD as a more general version of SI, compared to VBA. While the ability to persist and also die may be a universal trait of all lyfe in the universe, applying both VBA and CLAs to SI give a broader perspective for agnostically assessing the biogenicity of alien systems.

Moving to higher levels of complexity, one could experiment with the SI of real biological cells, and expose them to gene KO in different environments. The same interventions performed on the vesicles and droplets could be used, in order to compile a 'ladder of minimal to extreme interventions', which can be applied sequentially to a putative living sample under the SI approach to in situ biosignatures. Climbing this ladder should in general lead to the so-called semantic threshold for each system, the point at which further interventions are catastrophic to the viability of the agent in question. This series of experiments would allow the construction of a scale of SI values ranging from abiotic systems to progressively more complex biological cells. It would facilitate the development of standardised procedures and agnostic, experimentally grounded protocols applicable to the analysis of samples from future missions targeting Mars, Europa, Enceladus, or Titan.

Conclusions

In this paper, we proposed a general framework for assessing the semantic content of information called CLD. We present CLD as a basic SI measure that tracks all influences of a given informational intervention, overcoming the limitations of past approaches to SI, such as being confined to linguistic systems or measurements based upon an agent's viability. Such an intervention could be an erasing operation on an information-bearing structure such as a disk or genetic polymer. The system's states would then comprise two distributions: the un-intervened and the intervened, which evolve under dynamical laws, and combine deterministic and stochastic effects. The intervened distribution will either re-align itself with the original after some characteristic time, or may diverge indefinitely (at least until some point very far in the future, where both systems are essentially at equilibrium). The aim of CLD is to quantify the extent of divergence between the two sets of states. We proposed different metrics to be applied to this divergence: the instantaneous divergence of the two sets of phase space paths, the maximum measured divergence of the two sets over the period of observation, and the integrated divergence of the two sets over the period of observation.

Grounding SI in the framework of phase space trajectories and their divergences ensures that all possible causal influences of informational interventions can be captured. The method is therefore not limited to measuring the viability impacts of information, as is the case for recent VBA such as Kolchinsky and Wolpert (2018). While conceptually related to Lyapunov-based analyses in dynamical systems, CLD provides a more targeted and information-specific methodology. Rather than assessing divergence due to arbitrary state perturbations, CLD focuses on the consequences of specific informational manipulations, offering a generalisable foundation for quantifying semantic influence across systems.

We suggested a sequence of modeling and experimental studies that could provide benchmark assessments and refinements of both VBA and CLAs to SI, in the context of detecting life in the universe. Since we may encounter radically novel forms of lyfe that differ in composition and behaviour to terrestrial life, SI could offer a powerful, agnostic method for distinguishing living systems from the abiotic background, which itself could be highly complex.

Once developed, this general measure of SI could be of great benefit in biology, machine learning, economics, and computer science. It might also shed light on deep philosophical questions, such as the necessary and sufficient conditions for a system to show complex, emergent properties, and allow new explorations of a range of questions relevant to the ALife community.

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